ONLINE EXAM PORTAL ? TESTING EVIDENCE

Unit Test Report

Backend and frontend automated test execution

Repository: SE-Team-6/online_exam_portal

Branch: main **Commit:** 2914e5c55a79108e5b24fbb9bdd181fffbf119a1

Evidence date: 10 November 2026

Scope: Results and evidence collected for the current main checkout; no full-system sign-off implied.

Prepared from actual automated test output, browser validation output, screenshots, and the main-branch CI coverage artifact. Untested items are explicitly identified.

Execution summary

Suite	Command / environment	Actual result	Classification
Backend	`node --experimental-test-coverage --test` from `backend`; isolated MongoDB `oep_phase2_test`; Node v24.19.0	**27 passed / 27** 0 failed, 0 skipped	Unit and integration tests
Frontend	`npm --prefix frontend run test:coverage`; Vitest 4.1.12; V8 provider; jsdom	**12 passed / 12** 4 test files	Unit/component tests

MongoDB availability was confirmed by a successful ping before the backend integration tests. Frontend coverage output for this Phase 2 run was directed to a temporary directory outside the repository.

Backend test files and coverage

Test file	Cases	Areas exercised
`backend/test/app.test.js`	8	HTTP health response, JSON response, request IDs, malformed JSON, unsupported method, structured not-found, login contract and role denial. HTTP-level app checks.
`backend/test/attempt.integration.test.js`	6	MongoDB-backed attempt start/answer saving, ownership, idempotency, submission/grading, CSV validation, deadline enforcement and background auto-submit. Integration tests.
`backend/test/auth.integration.test.js`	2	MongoDB-backed failed-login lockout after five attempts and successful-login reset. Integration tests.
`backend/test/jwt.test.js`	6	JWT signing/expiry/signature and authentication/role middleware behavior. Unit tests.
`backend/test/password.test.js`	5	Salted password hashing, verification, malformed hashes and short-password rejection. Unit tests.

Frontend test files and coverage

Test file	Cases	Areas exercised
`frontend/src/components/ProtectedRoute.test.jsx`	3	Unauthenticated redirect, allowed-role access and student-to-student-dashboard redirect. Vitest/jsdom component tests.
`frontend/src/pages/QuestionBank.test.jsx`	1	CSV import UI sends parsed question payload and reports successful import. Vitest/jsdom component test.
`frontend/src/pages/Results.test.jsx`	3	Results display/percentile formatting, empty state and load-error display. Vitest/jsdom component tests.

Test file	Cases	Areas exercised
`frontend/src/utils/questionCsv.test.js`	5	CSV parsing of quoting/tags, BOM/blank rows and invalid headers/quotes/required data. Vitest unit tests.

Interpretation and boundaries

- Backend HTTP checks and MongoDB integrations exercise server/API behavior; they are not browser tests.
- Frontend Vitest tests use jsdom and mocked API calls in component tests; they are not real-browser system validation.
- Separate real-browser validation was performed and is documented in the System Validation Report.
- Passing these suites does not establish that all requirements or the entire system are validated.

Commit: 2914e5c55a79108e5b24fbb9bdd181fffbf119a1 ? Current branch: main.